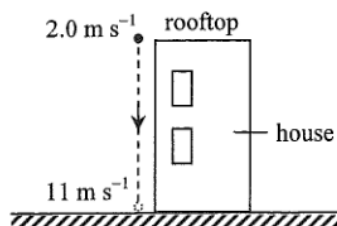


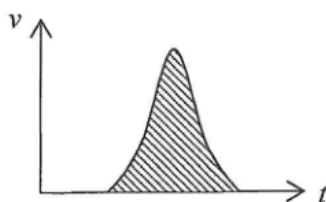
9.



A particle is projected vertically downward with an initial speed of  $2.0 \text{ m s}^{-1}$  from the rooftop of a house. The particle reaches the ground with a speed of  $11 \text{ m s}^{-1}$  as shown. Estimate the height of the house. Neglect air resistance. ( $g = 9.81 \text{ m s}^{-2}$ )

- A. 3.3 m
- B. 6.0 m
- C. 6.5 m
- D. 12 m

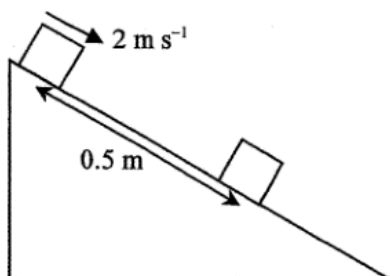
4.



What physical quantity does the shaded area represent in the above velocity-time ( $v-t$ ) graph?

- A. acceleration
- B. change of velocity
- C. momentum
- D. displacement

7. A block with initial speed  $2 \text{ m s}^{-1}$  slides down a rough inclined plane and stops after travelling a distance of  $0.5 \text{ m}$ . What is the deceleration of the block?



- A.  $1 \text{ m s}^{-2}$
- B.  $2 \text{ m s}^{-2}$
- C.  $4 \text{ m s}^{-2}$
- D. Answer cannot be found as the angle of inclination of the plane is not given.

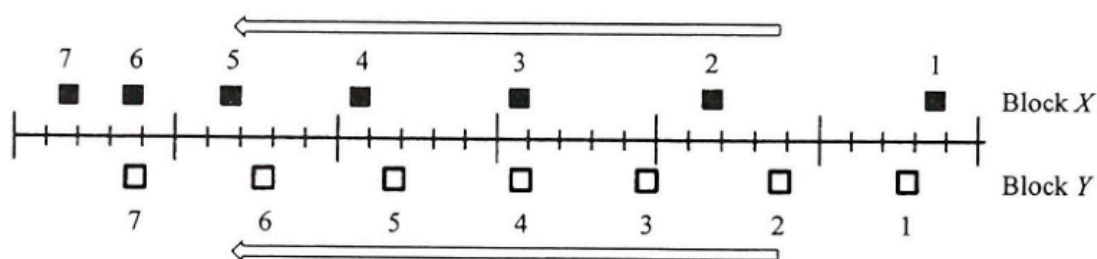
4. The speedometer of a car shown below indicates the car's



- A. instantaneous speed.
- B. instantaneous velocity.
- C. average speed of the whole journey.
- D. average velocity of the whole journey.

2018 Q7

7. Two blocks  $X$  and  $Y$  are moving towards the left. Their positions at successive instants (indicated by the numbers) of equal time intervals are shown below.



Do these two blocks ever have the same instantaneous speed ?

- A. Yes, at instant 3.
- B. Yes, at a moment between instants 4 and 5.
- C. Yes, at instant 6.
- D. No.

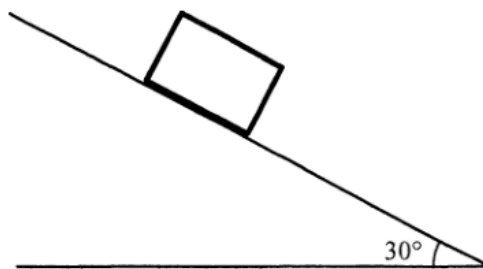
2017 Q5

5. Which of the following statements about the motion of any two objects is correct ?
- A. The object that takes a shorter time to complete the same path must have greater average speed.
  - B. The object that travels a greater distance in 1 s must have greater average velocity.
  - C. The object with greater velocity must have greater acceleration.
  - D. If the two objects have the same acceleration, they must be moving in the same direction.

2015 Q4

4. A particle travels at  $2.0 \text{ m s}^{-1}$  due east for 1.5 s and then travels at  $4.0 \text{ m s}^{-1}$  due north for 1.0 s. What is the magnitude of its average velocity for the whole journey ?
- A.  $2.0 \text{ m s}^{-1}$
  - B.  $2.8 \text{ m s}^{-1}$
  - C.  $3.0 \text{ m s}^{-1}$
  - D.  $5.0 \text{ m s}^{-1}$

6. A block is released from rest on an inclined plane as shown. The inclined plane makes an angle of  $30^\circ$  to the horizontal. The block moves with uniform acceleration, and travels a distance of 1 m in the first 3 s. Determine the acceleration of the block.



- A.  $0.22 \text{ m s}^{-2}$   
 B.  $0.33 \text{ m s}^{-2}$   
 C.  $4.91 \text{ m s}^{-2}$   
 D. Cannot be determined as the frictional force acting on the block is unknown.

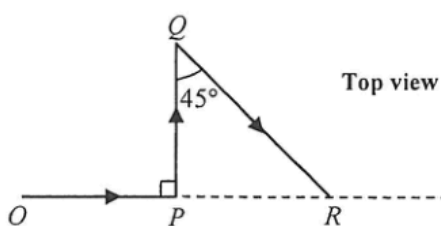
6. A particle is thrown vertically upward and its path is shown below.  $H$  is the highest point that the particle reached. Which of the following statements about the particle is/are correct? Neglect air resistance.

- (1) Its acceleration at  $M$  is upward.  
 (2) Its acceleration at  $H$  is zero.  
 (3) Its acceleration at  $N$  is downward.

- A. (1) only  
 B. (3) only  
 C. (2) and (3) only  
 D. (1), (2) and (3)



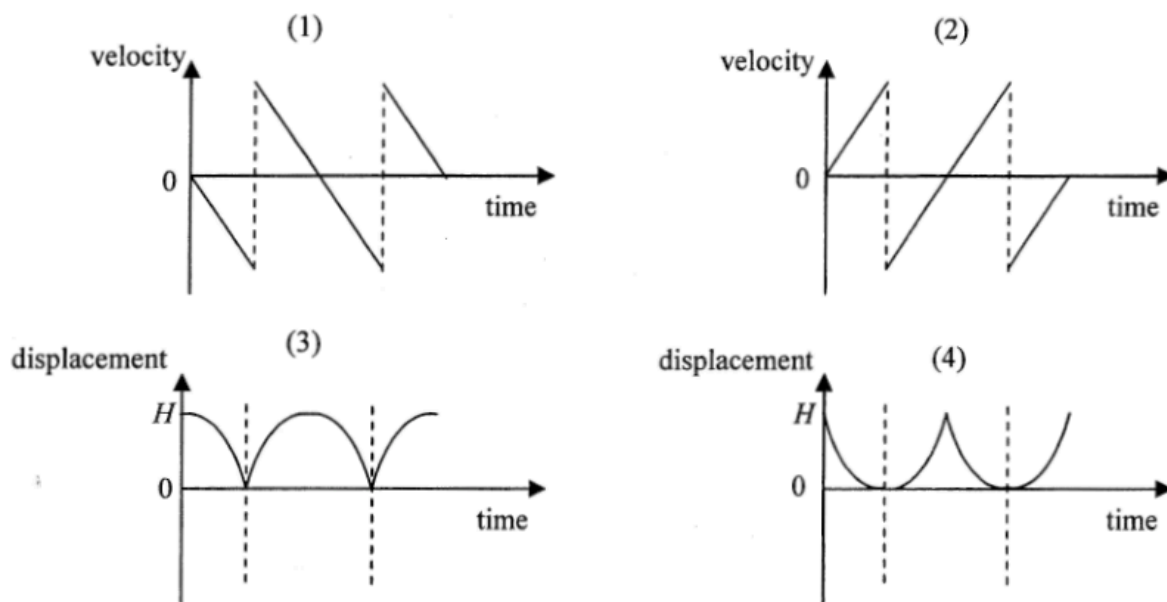
4.



A car takes 8 minutes to travel along a path  $OPQR$  on a horizontal surface as shown. Given that  $OP = PQ = 2 \text{ km}$ , find the magnitude of the average velocity of the car in this journey.

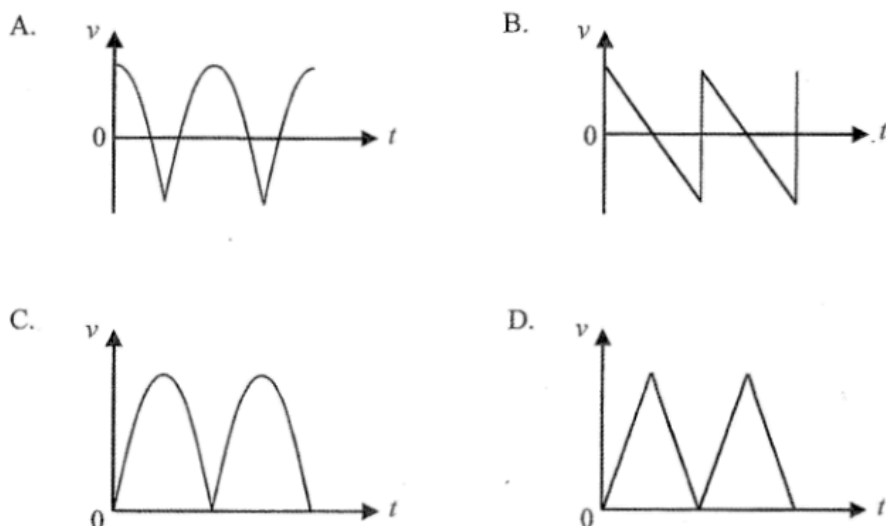
- A.  $30 \text{ km h}^{-1}$   
 B.  $36 \text{ km h}^{-1}$   
 C.  $41 \text{ km h}^{-1}$   
 D.  $51 \text{ km h}^{-1}$

7. Which of the following graphs (velocity-time and displacement-time) best represent the motion of a ball falling from rest under gravity at a height  $H$  and bouncing back from the ground two times? Assume that the collision with the ground is perfectly elastic and neglect air resistance. (Downward measurement is taken to be negative.)



- A. (1) and (3) only  
 B. (1) and (4) only  
 C. (2) and (3) only  
 D. (2) and (4) only

7. A rubber ball bounces vertically up and down from the ground. Which graph best shows the variation of its velocity  $v$  with time  $t$  if the collisions are elastic? Neglect air resistance.



4. Which of the following are vector quantities ?

- (1) acceleration
- (2) momentum
- (3) work done

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2018 Q9

9. A particle moving along a straight line with speed  $0.5 \text{ m s}^{-1}$  changes its direction of motion within a period of  $0.2 \text{ s}$  and subsequently travels with the same speed in the opposite direction. Find the magnitude of its average acceleration during this  $0.2 \text{ s}$ .

- A.  $5 \text{ m s}^{-2}$
- B.  $2.5 \text{ m s}^{-2}$
- C.  $0 \text{ m s}^{-2}$
- D. its magnitude depends on the mass of the particle

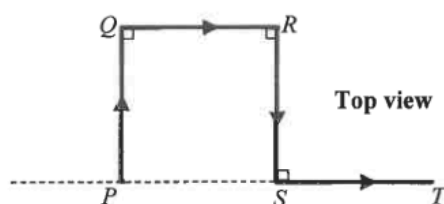
2018 Q10

10. A train departs from one station and stops at the next station after travelling  $1.2 \text{ km}$  along a straight line. The maximum acceleration and deceleration of the train are both  $5 \text{ m s}^{-2}$  and the highest speed of the train is limited to  $20 \text{ m s}^{-1}$ . Find the shortest time taken for this trip.

- A.  $56 \text{ s}$
- B.  $58 \text{ s}$
- C.  $62 \text{ s}$
- D.  $64 \text{ s}$

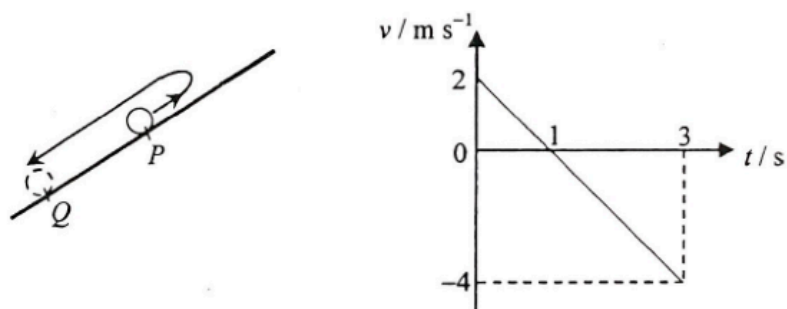
2022 Q4

4. A car travels with constant speed  $v$  along a horizontal road  $PQRST$  with four sections of equal length as shown. Determine  $v$  if the car's average velocity for the journey  $PQRST$  is  $20 \text{ km h}^{-1}$  in magnitude.



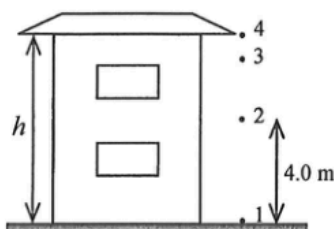
- A.  $10 \text{ km h}^{-1}$
- B.  $20 \text{ km h}^{-1}$
- C.  $40 \text{ km h}^{-1}$
- D. It cannot be found as the length of each section is unknown.

11. A bead is projected up along a smooth incline at point  $P$  at time  $t = 0$ . After reaching the highest point, the bead then travels downwards and passes point  $Q$  at  $t = 3$  s as shown. The velocity-time ( $v$ - $t$ ) relationship of the bead is shown in the graph below. Find the separation  $PQ$  along the incline.



- A. 2 m  
B. 3 m  
C. 4 m  
D. 5 m

4.



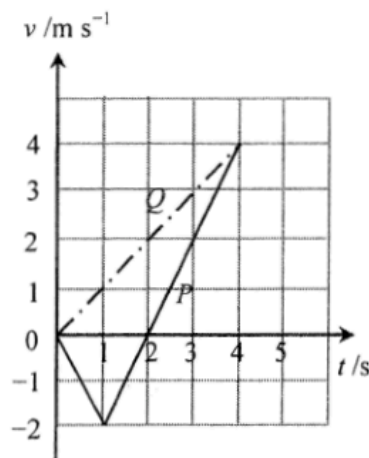
Water drips regularly at a constant rate from a building's roof, which is at a height  $h$  above the ground. The droplets fall from rest such that the first droplet reaches the ground just when the fourth droplet leaves the roof as shown. If the second droplet is 4.0 m above the ground at this instant, find  $h$ . Neglect air resistance.

- A. 5.8 m  
B. 6.4 m  
C. 7.2 m  
D. 7.8 m

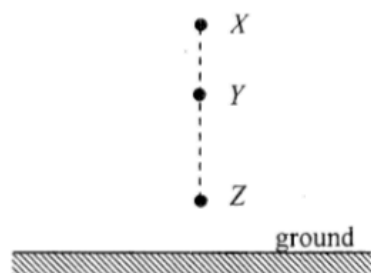
11. Two particles  $P$  and  $Q$  start from the same position and travel along the same straight line. The figure shows the velocity-time ( $v$ - $t$ ) graph for  $P$  and  $Q$ . Which of the following descriptions about their motion is/are correct?

- (1) At  $t = 1$  s,  $P$  changes its direction of motion.  
(2) At  $t = 2$  s, the separation between  $P$  and  $Q$  is 4 m.  
(3) At  $t = 4$  s,  $P$  and  $Q$  meet each other.

- A. (1) only  
B. (2) only  
C. (1) and (3) only  
D. (2) and (3) only



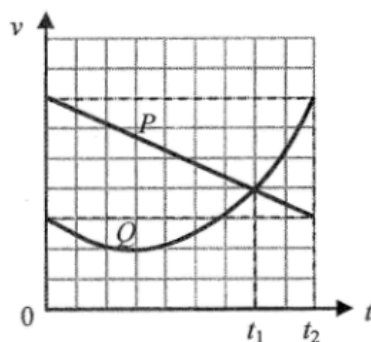
8.



A particle is released from rest at  $X$  as shown. It takes time  $t_1$  to fall from  $X$  to  $Y$  and time  $t_2$  to fall from  $Y$  to  $Z$ . If  $XY : YZ = 9 : 16$ , find  $t_1 : t_2$ . Neglect air resistance.

- A. 2 : 3
- B. 3 : 4
- C. 4 : 3
- D. 3 : 2

9.



The figure shows the velocity-time ( $v-t$ ) graph of two cars  $P$  and  $Q$  travelling along the same straight road. At  $t = 0$ , the cars are at the same position. Which deductions about the cars between  $t = 0$  and  $t = t_2$  are correct?

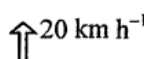
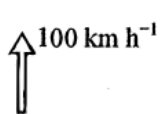
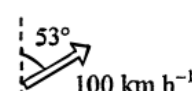
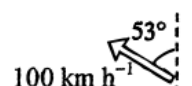
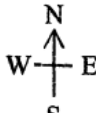
- (1)  $P$  and  $Q$  are always travelling in the same direction.
- (2) At  $t = t_1$ , the separation between  $P$  and  $Q$  is at a maximum.
- (3) At  $t = t_2$ ,  $Q$  lags behind  $P$ .

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

5. A particle is moving along a straight line with uniform acceleration. It takes 4 s to travel a distance of 36 m and then 2 s to travel the next 36 m. What is its acceleration?

- A.  $2.5 \text{ m s}^{-2}$
- B.  $3.0 \text{ m s}^{-2}$
- C.  $4.0 \text{ m s}^{-2}$
- D.  $4.5 \text{ m s}^{-2}$

5. A car travelling at  $80 \text{ km h}^{-1}$  due east changes direction and travels at  $60 \text{ km h}^{-1}$  due north. Which diagram represents the change in velocity of the car?

- A.   $20 \text{ km h}^{-1}$
- B.   $100 \text{ km h}^{-1}$
- C.   $100 \text{ km h}^{-1}$
- D.   $100 \text{ km h}^{-1}$
- 

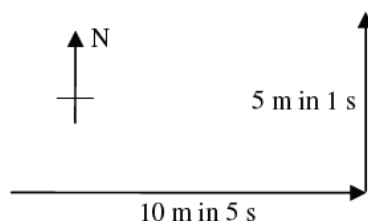
7. At time  $t = 0$ , a small sphere is projected up along a smooth incline with a certain initial velocity. It travels a distance  $L$  and becomes momentarily at rest after a time  $2T$ . The corresponding velocity-time graph is shown below.



What is the distance travelled by the sphere from  $t = 0$  to  $t = T$ ?

- A.  $\frac{1}{4}L$
- B.  $\frac{1}{2}L$
- C.  $\frac{3}{4}L$
- D.  $\frac{4}{5}L$

6.



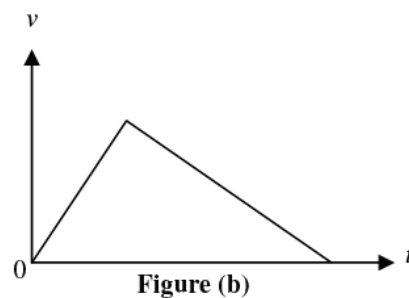
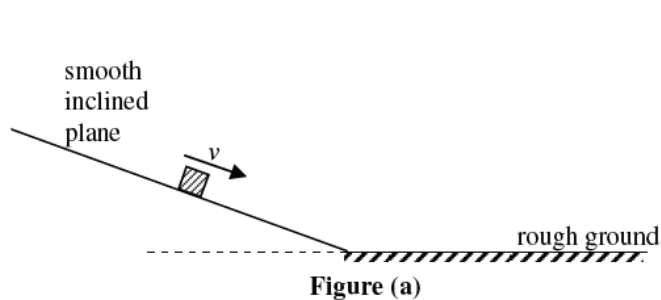
A toy car travelled due east for 10 m in 5 s, then immediately turned north and travelled 5 m for 1 s. What was the average speed of the car?

- A.  $1.9 \text{ m s}^{-1}$
- B.  $2.2 \text{ m s}^{-1}$
- C.  $2.5 \text{ m s}^{-1}$
- D.  $3.5 \text{ m s}^{-1}$

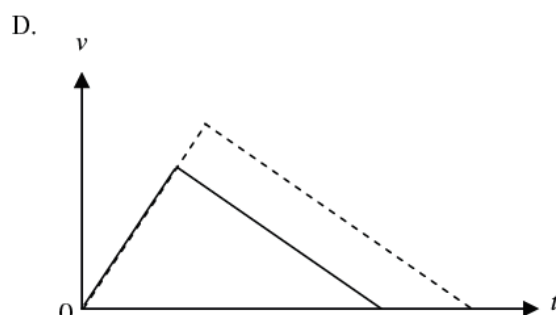
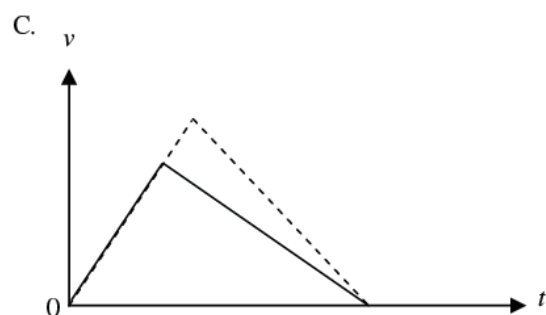
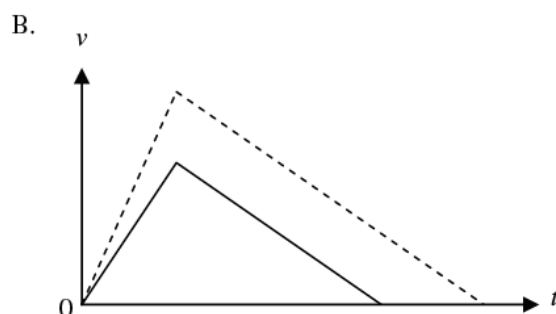
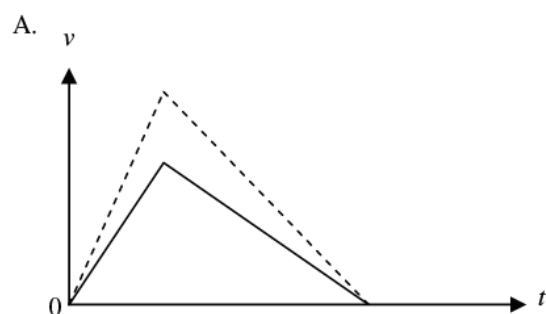


7. A stone falls from rest. Neglecting air resistance, the ratio of the distance travelled by the stone in the 1<sup>st</sup> second to that travelled in the 2<sup>nd</sup> second is
- A. 1 : 1  
 B. 1 : 2  
 C. 1 : 3  
 D. 1 : 4

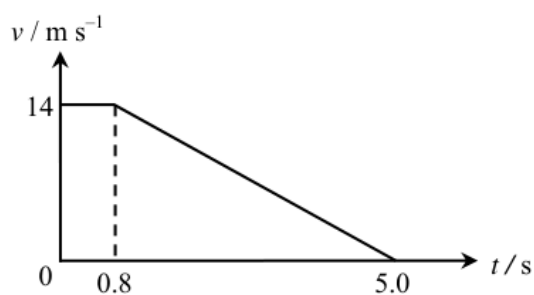
10.



As shown in Figure (a), a block slides down along a smooth inclined plane from rest. The corresponding speed-time graph of its motion is shown in Figure (b). Which of the following speed-time graphs (in dotted lines) best represents the motion of the block if it is released at a higher position on the plane instead? Assume that the friction between the ground and the block remains unchanged.



7. Patrick is driving along a straight horizontal road. At time  $t = 0$ , he observes that an accident has happened. He then applies the brakes to stop his car with uniform deceleration. The graph shows the variation of the speed of the car with time.



Find the distance travelled by the car from time  $t = 0$  to 5.0 s.

- A. 29.4 m
- B. 40.6 m
- C. 46.2 m
- D. 81.2 m

12. Two small identical objects  $P$  and  $Q$  are released from rest from the top of a building 80 m above the ground.  $Q$  is released 1 s after  $P$ . Neglecting air resistance, what is the maximum vertical separation between  $P$  and  $Q$  in the air?

- A. 5 m
- B. 10 m
- C. 35 m
- D. 45 m